

Hypothesis-Discriminating Observation Campaigns for Distributed Space Observatories

Sep 24, 2026

Reader's note

The paper this project should produce is about one precise gap: a constellation that must plan a multi-step, multi-spacecraft observation campaign to tell competing explanations of a fading event apart, while staying honest that the true explanation may not be in its hypothesis set. OrbitalNet is the substrate that makes the campaign distributed; it is not the contribution.

Three status labels are used throughout. **Verified** means a source was opened while writing this. **Draft claim** means reasoning that you must check against the literature yourself. **Expected** means a result the experiments are designed to test; nothing in this document has been demonstrated yet.

The work is structured as the master prompt asked: landscape of doors (Part I), comparison (Part II), deep prior art on the top three (Part III), the paper (Part IV), venues (Part V), and a week-by-week plan from the simulator you already have (Part VI). Every numbered reference must be re-read by you before submission; treat my summaries as leads, not citations.

Part I — Research landscape: twelve doors

Twelve naive questions were each followed down to a bottleneck, checked against prior art, and assigned a novelty class. Four doors lead somewhere real; five lead to solved or weakly novel territory; three are useful only as supporting pieces. The full descent is written out only for the three that survive (Part III), so this table stays readable.

Novelty classes follow your methodology: **A** already solved, **B** known problem + known method, **C** known problem + incomplete solution, **D** known method + structurally different problem, **E** existing components + previously uncombined scientific objective, **F** fundamental open question.

#	Naive question	Descent (answer → hidden assumption → real bottleneck)	Closest prior art	Class	Candidate research question
D1	Why don't we just look again?	We do: brokers rank events for follow-up. Assumption: follow-up value is "how uncertain is the classifier". Bottleneck: ranking says <i>which object</i> , not <i>which measurement from which instrument, when</i> .	RESSPECT and Fink active learning pick objects to label a training set; Astudillo et al. pick which objects deserve a spectrum	C	For one live event, which instrument/epoch pair most separates its surviving explanations?
D2	If we don't know what it is, what should we measure next?	Depends on which explanations remain. Assumption: a fixed follow-up recipe per class. Bottleneck: the recipe is chosen before the class is known.	JPL POISE (Earth science, hypothesis-discriminating re-tasking); Science Hypothesis Maps and SHM-POMDP (rover geology)	C	Can sequential Bayesian model discrimination plan a follow-up campaign for a transient across heterogeneous space instruments?
D3	What if none of our explanations are right?	Classifiers force a label. Assumption: truth is in the hypothesis set. Bottleneck: expected-information-gain planners fail badly when the model is wrong, and they spend the budget confirming a wrong answer.	Misspecification-robust experimental design (robust EIG with a misspecification indicator; EGIG/EDI; GBOED) — all in ML/statistics, none in observation campaigns	D	Can a campaign planner reserve measurements that test its own model space, and switch from discrimination to preservation when the model fails?

#	Naive question	Descent (answer → hidden assumption → real bottleneck)	Closest prior art	Class	Candidate research question
D4	Why can't two spacecraft act as one instrument?	They can, for a known objective. Assumption: the objective is fixed and shared. Bottleneck: when the objective is "reduce uncertainty about <i>this</i> event", each node's value depends on beliefs it may not have received yet.	NASA DSA on Starling (distributed allocation for ionospheric TEC); consensus/auction constellation scheduling	E	How should intermittently connected spacecraft allocate discriminating measurements when their beliefs are stale and unequal?
D5	What happens when the universe does something only once?	The event fades; later data cannot replace it. Assumption: the next measurement can wait for the "best" one. Bottleneck: value of a measurement decays at a physics-set rate that differs by hypothesis.	Target-of-opportunity urgency ranking; deadline-aware scheduling generally	C	How should a planner trade discriminating power against the closing window of a fading source?
D6	Why do we throw data away before we understand it?	Storage and downlink are finite. Assumption: value can be judged at capture. Bottleneck: data that matters only under an unmodeled hypothesis is discarded first.	Onboard prioritization and summarization (e.g. ASE/EO-1 lineage, JPL onboard science)	C	Can retention priority be tied to the probability that the current model space is inadequate?

#	Naive question	Descent (answer → hidden assumption → real bottleneck)	Closest prior art	Class	Candidate research question
D7	Why can't we see the far side of things?	Geometry limits what one viewpoint measures. Parallax from separated spacecraft already done for microlensing.	Space-based microlensing parallax campaigns	B	Mostly solved; weak as a main line.
D8	What does "now" mean across spacecraft?	Arrival-time differences localize events; the Interplanetary Network does this for gamma-ray bursts.	IPN triangulation	A	Solved; drop.
D9	Why is a detection a yes/no?	Alerts carry scores; probabilistic alerts exist.	Broker classification streams	B	Weak; supporting piece only.
D10	How do two spacecraft know they saw the same thing?	Cross-matching under latency and positional error.	Standard association and multi-target tracking	B	Supporting piece only.
D11	Why must a telescope be told what is interesting?	Anomaly detection for transients is active.	Real-time transient anomaly detection (Villar et al.; Muthukrishna et al.)	B/C	Detection exists; <i>what to do after flagging</i> is the open part — folds into D3.
D12	Why can't the constellation ask itself for help?	Request/offer protocols (contract net, auctions) exist.	Consensus-based bundle and market allocation	B	Mechanism only; becomes the transport for D4.

The pattern across the table matters more than any single row. Every lower-stack capability (detect, rank, allocate, retarget) has prior art. What recurs as unresolved is the layer where a **distributed** system must reason about **which explanation is true**, under a **closing window**, while allowing that **its explanations may all be wrong**. D2, D3, D4 and D5 are four views of that one layer.

Part II — Comparing the directions

The strongest paper merges D2 (hypothesis discrimination) with D5 (irreversibility) as its core, adds D3 (model inadequacy) as its distinguishing feature, and uses D4 (distribution) as the setting that makes it hard. No arbitrary scores are used; each criterion below names the door that wins it and why.

Criterion from your brief	Strongest door	Reason
Clearest unresolved problem	D3	Robust experimental design exists in statistics but has not been tied to time-critical, multi-instrument observation. The gap is narrow and nameable.
Strongest scientific consequence	D5	A missed early epoch of a kilonova or shock-cooling supernova is physically unrecoverable. The loss is concrete.
Strongest distinction from prior work	D3 + D4 together	POISE and SHM-POMDP reason over hypotheses but assume the set is complete and a single planner. DSA distributes allocation but has no hypothesis state. Nobody sits at the intersection.
Most convincing experimental path	D2 + D5	Injected synthetic transients give ground truth; baselines are easy to define; metrics are standard.
Strongest connection to OrbitalNet	D4	Contact windows, stale beliefs and node loss are exactly what your orbital network layer already models.
Greatest potential as a real contribution	D2+D3+D5 in the D4 setting	Each alone is incremental. The combination creates a testable capability none of the prior systems claims.

Risk of the merge. Four ideas in one paper can read as unfocused. The fix is a single headline claim, with the rest as enabling pieces: *a distributed planner that keeps an explicit “none of the above” hypothesis recovers more of the physically unrecoverable information about fading transients than closed-world or reactive planners under the same constraints.*

Doors rejected and why. D7 and D8 are solved. D9, D10 and D12 are engineering components you will need, but no reviewer would accept them as the contribution. D6 is promising but is better as a follow-up paper once D3 exists, because its value depends on the inadequacy signal D3 produces. D11 is absorbed into D3.

Part III — Deep prior art and the gap

The gap survives, but only in a narrow form: no located system plans a *multi-instrument, distributed, deadline-bound* observation campaign whose objective is discriminating hypotheses about a single astrophysical event *while testing whether its hypothesis set is adequate*. Each of those qualifiers is covered separately somewhere; the conjunction is not. That is class E, not class F. Do not claim more.

The full descent for the surviving line

1. **Simple question.** A new source brightens in one image. What should we do?
2. **First answer.** Observe it again with more instruments.
3. **Deeper question.** Which instrument, which band, when? Every choice costs time the source may not have.
4. **Hidden assumption.** Follow-up recipes assume you know the class. Early on, a shock-cooling supernova, a kilonova, a GRB afterglow and a stellar flare can look alike in two optical bands.
5. **Real bottleneck.** The best next measurement depends on which explanations are still alive, and that changes after every measurement.
6. **Obvious solution.** Maximize expected information gain over the classes (standard Bayesian experimental design).
7. **Why it isn't enough.** (a) It assumes the true class is in the set; when it isn't, it confidently confirms a wrong answer. (b) It ignores that the value of a UV epoch at hour 6 is zero at hour 40. (c) It assumes one planner with current beliefs; a constellation has several planners with stale ones.

8. **Information lost.** Early colour evolution and spectra of fast transients; raw data for events no model explains.
9. **Cross-domain pattern.** Differential diagnosis in emergency medicine: tests are chosen to separate the remaining diagnoses, with "something else" always kept open, under a clock. Robust experimental design supplies the maths for (a); deadline-aware scheduling for (b); decentralized information gathering for (c).
10. **Return to space.** Instruments on different spacecraft are different tests; contact windows decide who knows what.
11. **New capability.** A constellation that runs a differential diagnosis on a fading sky event and knows when to stop diagnosing and start preserving.

Closest prior work, examined

Work	What it does	Assumes	Does not do	Delta for this paper
JPL POISE	Chooses observations that best discriminate competing hypotheses and re-tasks assets under cost and feasibility (Earth science, e.g. cyclone rapid intensification)	Hypotheses given by domain models; Earth-observing assets	Astrophysical transients; explicit unmodeled hypothesis; decentralized planning under intermittent links (verify in full papers)	Closest conceptual ancestor. Cite prominently and position as extending its idea to open-world, distributed, fading-source astronomy.
SHM-POMDP (Aug 2026) and Science Hypothesis Maps (Candela et al., IROS 2017)	Belief-space planning over science hypotheses for a rover; outperforms information-theoretic baselines on hyperspectral geology	Single agent; static target; closed hypothesis structure	Multi-agent; time-decaying targets; model inadequacy	Shows the community values hypothesis-driven planning. You differ on agents, time and openness.

Work	What it does	Assumes	Does not do	Delta for this paper
NASA DSA on Starling	Four spacecraft share perspectives and allocate ionospheric measurements without ground direction; survive member loss	Objective fixed (capture ionospheric phenomena); no hypothesis state	Hypothesis discrimination; astronomy	Proves distributed allocation flies. Your objective is epistemic, not coverage.
Astudillo et al., AJ 2020 (verify authors)	Information-theoretic choice of which objects get a spectrum	One follow-up type; population-level; no timing	Sequential, multi-instrument, per-event campaigns	Nearest astronomy precedent for information-theoretic follow-up.
RESSPECT and Fink real-time active learning	Choose spectroscopic targets to improve classifier training sets	Goal is a better classifier for a population	Per-event discrimination; misspecification	Different objective. Say so explicitly; reviewers will ask.
NMMA-Astro-COLIBRI (2026)	Fits early photometry to competing models and reports Bayes factors; recovered a IIb mis-flagged as a kilonova	Model comparison after data arrive	Choosing the next measurement	Excellent motivating case and possible forward-model source.
Robust BED under misspecification ; EGIG/EDI ; GBOED	Design experiments that remain informative, or detect misspecification, when the model is wrong	Single experimenter; no hard deadlines; no distributed assets	Space observation	You borrow the mechanism (auxiliary model + misspecification indicator). Your delta is the operational setting and the switch to preservation.
GRANDMA, TOM Toolkit/AEON	Human-coordinated global follow-up networks for transients	Humans decide the campaign	Autonomous hypothesis-driven campaign logic	Your planner is what such networks could call. Realistic users.

Work	What it does	Assumes	Does not do	Delta for this paper
On-orbit space AI survey (2026)	Reviews decentralized and auction-based constellation scheduling under intermittent connectivity	Task objectives given	Epistemic objectives	Use to justify your coordination baseline.

Kill attempts

- *"POISE already does this."* Partly. POISE discriminates hypotheses and re-tasks assets. Your defensible delta is the unmodeled hypothesis, the fading-source deadline and decentralized stale beliefs. If a POISE paper shows any of these, drop that claim. **Read every POISE publication before writing.**
- *"This is just Bayesian experimental design."* The maths is borrowed on purpose. The contribution is showing which parts break in this setting and measuring the capability gained. Frame it as systems science, not new theory.
- *"Only latency and bandwidth improve."* No: the headline metric is correct identification and recovered early-time physics of events closed-world planners get wrong. That is knowledge, not throughput.
- *"Simulation only."* True for the planning claim. Mitigated by physics-grounded forward models, real light-curve templates, withheld classes, and a hardware testbed that reproduces the communication constraints.

Part IV — The paper

Title

Open-World, Deadline-Aware Hypothesis Discrimination for Distributed Space Observatories: Planning Follow-Up Campaigns for Fading Transients Under Intermittent Connectivity

Short form for talks: *Differential Diagnosis for the Sky.*

Abstract (draft; results are hypotheses until run)

Fast optical transients such as kilonovae, shock-cooling supernovae, relativistic afterglows and stellar flares are often indistinguishable in their first hours, yet the measurements that separate them lose value within hours to days as the sources fade. Current follow-up is organised by recipes chosen per assumed class, by classifier-uncertainty ranking of objects, or by distributed allocation toward fixed objectives. None of these plans a sequence of heterogeneous measurements whose purpose is to discriminate competing physical explanations of one event before the discriminating signal disappears, and none accounts for the possibility that the true explanation lies outside the hypothesis set. We formulate follow-up as open-world sequential model discrimination with time-decaying measurement value, executed by spacecraft that hold unequal, intermittently synchronised beliefs. The planner maintains a posterior over K physical models plus an explicit unmodeled alternative represented by a flexible auxiliary model, selects measurements by a utility combining expected discrimination, expected evidence of model inadequacy and opportunity decay, and switches from discrimination to evidence preservation when inadequacy becomes probable. Allocation across spacecraft uses a marginal-gain auction that tolerates stale beliefs. We evaluate on a simulated heterogeneous low-Earth-orbit constellation with physically grounded visibility, slew, sensitivity and contact constraints, using injected transients from published light-curve models and withheld classes as ground-truth anomalies, and replicate the communication layer on an ESP32 hardware-in-the-loop testbed. We test whether the planner reaches correct identification earlier, recovers more early-time information, and flags withheld classes more reliably than reactive, uncertainty-sampling and closed-world information-gain baselines under identical budgets.

Research problem

When an unexplained transient appears, a constellation of heterogeneous instruments must decide, repeatedly and without waiting for the ground, which spacecraft should take which measurement next. The right choice depends on which explanations remain plausible, how fast each explanation predicts the discriminating signal will vanish, what other spacecraft are already doing, and whether any existing explanation fits at all.

Motivation

The measurement that separates two hypotheses is often the one that is available only briefly: early ultraviolet colour, the first spectrum before a photosphere cools, a near-infrared epoch before a kilonova fades. A wrong or late choice is not a delay; it is a

permanent loss. Wide-field surveys now produce far more candidates than follow-up can serve, which makes the choice of measurement, not just of target, the binding constraint. Real cases exist: automated pipelines mis-flagged a Type IIb supernova as a kilonova candidate for days before spectroscopic confirmation.

Research gap

Hypothesis-discriminating observation exists for Earth science and rover geology with a closed hypothesis set and a single planner. Distributed spacecraft allocation exists for fixed coverage objectives. Misspecification-robust experimental design exists for single experimenters without deadlines. Astronomical follow-up optimisation targets training-set construction or one-shot spectroscopic selection. No located work combines all four in one campaign planner and measures what the combination recovers.

Research question

Can a distributed, resource-constrained space observatory plan sequential multi-instrument measurements that discriminate competing physical explanations of a fading transient, and recognise when none of them applies, better than reactive and closed-world planners under identical physical and communication constraints?

Hypotheses (falsifiable)

- **H1 (discrimination).** The proposed planner reaches 95% posterior on the true modeled class in less median time than all baselines, with no more measurements.
- **H2 (irreversibility).** Removing the opportunity-decay term reduces recovered early-time information for fast classes (kilonova, shock cooling) more than for slow classes.
- **H3 (open world).** On withheld classes, the planner's inadequacy probability separates unmodeled from modeled events better than classifier-entropy thresholds, at a matched false-alarm rate.
- **H4 (distribution).** Performance degrades gracefully as inter-satellite contact fraction falls; the stale-belief auction keeps a measurable fraction of centralized-oracle performance at contact fractions where independent greedy planners duplicate measurements.

Each hypothesis names the result that would refute it. Report refutations honestly; a refuted H4 is still a publishable finding.

Contributions

1. A formulation of transient follow-up as open-world sequential model discrimination with hypothesis-dependent measurement decay.
2. A decentralized allocation rule for epistemic objectives under intermittent inter-satellite links.
3. A discrimination-to-preservation switch driven by model-inadequacy evidence.
4. An open simulator and benchmark with withheld-class tests, plus a hardware-in-the-loop communication testbed.

Paper: architecture and formulation

OrbitalNet supplies three things the planner needs and nothing more: who can see the target when, who can talk to whom when, and a message layer that survives link gaps. The contribution sits in the belief and utility layers above it.

flowchart TD

```

A[Orbital layer<br/>SGP4 visibility, slew, eclipse] --> C
B[Network layer<br/>ISL contact graph, DTN queue] --> D
C[Feasible action set<br/>per spacecraft, per time] --> E
D[Belief exchange<br/>posterior + commitments] --> E
E[Utility engine<br/>discrimination + inadequacy + decay] --> F
F[Marginal-gain auction<br/>stale-belief tolerant] --> G
G[Execute measurement] --> H
H[Local Bayesian update] --> D
H --> I{P unmodeled > tau?}
I -->|yes| J[Preservation mode<br/>broad coverage, keep raw data]
I -->|no| E

```

Each spacecraft runs the whole loop locally; the auction and belief exchange happen only when the contact graph allows it.

Hypothesis space

The event has an unknown class h drawn from K physical models plus one unmodeled alternative h_o . Each modeled class k has parameters θ_k (explosion time, distance, ejecta mass, temperature, and so on) and a forward model predicting the flux an instrument would measure.

$$\mathcal{H} = \{h_1, \dots, h_K\} \cup \{h_0\}, \quad y \mid h_k, \theta_k, a \sim \mathcal{N}\big(f_k(\theta_k, \lambda_a, t_a), \sigma_a^2(f_k)\big)$$

The unmodeled alternative uses a deliberately flexible auxiliary model, for example a Gaussian process over time and wavelength with broad priors. It never wins by fitting a known class well; it wins only when all known classes fit badly.

$$y \mid h_0, a \sim \mathcal{GP}\big(m(t, \lambda), k_{\text{broad}}((t, \lambda), (t', \lambda'))\big)$$

Actions and feasibility

An action $a = (\text{spacecraft } i, \text{instrument } m, \text{band or spectral mode } \lambda, \text{start time } t, \text{exposure } \Delta)$. It is feasible only if the orbital layer reports the target visible and outside Sun, Earth-limb and South Atlantic Anomaly constraints, the slew fits, and power and storage budgets hold. The noise σ_a comes from the instrument's limiting-magnitude model, so a faded source yields an uninformative measurement automatically.

Utility

The score of a candidate measurement combines three terms, divided by its resource cost $c(a)$.

$$U(a \mid b_t) = \frac{I(H_{1:K}; Y_a \mid b_t) + \beta I(Z; Y_a \mid b_t) + \gamma D(a, b_t)}{c(a)}$$

- The first term is the mutual information between the measurement and the class label: how much it separates the remaining explanations.
- Z is a binary indicator of whether the truth is h_0 . The second term is how much the measurement would reveal about model adequacy. This follows the robust-design idea of adding a misspecification indicator.
- The third term is opportunity decay: the expected loss if this measurement is postponed to the next feasible slot, averaged over the posterior. Classes that fade fast make early measurements urgent only when those classes are still plausible.

$$D(a, b_t) = \sum_k b_t(h_k) \big[I_k(Y_a) - I_k(Y_{a'}) \big]_+, \quad a' = \text{same measurement at next feasible time}$$

Multi-spacecraft allocation

For a fixed class variable and measurements that are conditionally independent given the class, mutual information is monotone submodular, so greedy selection of a batch is within a factor $(1 - 1/e)$ of optimal. That guarantee holds only for a centralized planner with shared beliefs. Your auction lets each spacecraft bid its marginal gain given the commitments it has heard about. Stale beliefs break the guarantee; **measuring how much is result H4**, not something to hide.

Mode switch

If the posterior on h_0 exceeds a threshold τ for two consecutive updates, the utility changes: discrimination terms are dropped and the planner maximizes coverage in time and wavelength of the auxiliary model, and raises onboard retention priority for all raw data on this event. τ is set on a validation split to hit a chosen false-alarm rate on modeled classes.

Belief update and communication

Each spacecraft keeps a particle or grid posterior per class. Messages carry per-class log-evidence and a compact list of committed future measurements, not raw data. Merging two beliefs that share earlier measurements must avoid double counting; tagging each measurement with a unique ID and summing only unseen evidence handles this exactly for discrete classes.

Paper: experimental design

Every strategy sees the same injected events, the same orbits, the same instrument budgets and the same contact schedule; only the decision rule changes. That pairing is what turns your simulator into evidence.

Simulated universe

Component	What to use	Why
Modeled classes (K = 4)	Kilonova, shock-cooling / Type IIb early phase, GRB afterglow (broken power law), M-dwarf flare	All plausible early confusions; published analytic or template models exist

Component	What to use	Why
Withheld classes (truth = h_0)	A fast blue optical transient (AT2018cow-like), a tidal disruption event, and one synthetic "nonsense" class with physically odd colour evolution	Ground truth for the open-world test; the planner never sees these models
Forward models	Use published light-curve codes or templates (for example sncosmo templates, kilonova grids, flare templates); cite each source	Physics grounding; reviewers will check this first
Distances and rates	Sample distance and host extinction from realistic ranges; weight classes by rough relative rates	Avoid a toy balanced problem

Simulated constellation

Six to eight LEO spacecraft from your OrbitalNet propagator, mixed payloads: two UV imagers, two optical multi-band imagers, one near-infrared imager, one low-resolution spectrograph, optionally one X-ray monitor. Each instrument has a limiting-magnitude-versus-exposure curve, slew rate and duty-cycle limits. Visibility uses SGP4 with Earth occultation, Sun avoidance and SAA exclusion. Inter-satellite links follow line-of-sight and range, sampled into a contact schedule.

Strategies compared

Label	Strategy	Role
B0	Fixed recipe: two optical bands every N hours, spectrum when brighter than a threshold	Human-style practice
B1	Your current reactive dispatcher (nearest capable spacecraft responds)	Shows what the existing project does

Label	Strategy	Role
B2	Classifier-entropy sampling: measure whatever reduces predicted-class entropy, no physics likelihoods	Active-learning-style baseline
B3	Closed-world expected information gain, centralized, no decay term, no h_0	POISE-style baseline; the key comparison
P	Full method: h_0 + decay + stale-belief auction	Proposed
O	Centralized, perfect communication, full method	Upper bound

Metrics

- **Time to identification:** hours until posterior on the true class exceeds 0.95; censored at the event's detectability limit.
- **Accuracy at deadline:** correct top class and log score at a fixed horizon (for example 72 h).
- **Early-time information recovered:** posterior contraction of each class's physical parameters (for example ejecta mass, explosion time) relative to prior; this is the knowledge metric.
- **Open-world detection:** area under the ROC curve of $P(h_0)$ separating withheld from modeled events; false-alarm rate on modeled events at the chosen τ .
- **Preservation:** for withheld events, fraction of the event's detectable lifetime covered in at least three wavelength regions.
- **Cost:** measurements used, slew time, messages sent, bytes exchanged.

Scenarios

1. Nominal: full constellation, contact fraction as propagated.
2. Communication sweep: artificially thin contacts from 100% down to 10% of the propagated schedule.
3. Node loss: remove one or two spacecraft mid-campaign.
4. Crowding: several events at once competing for the same instruments.
5. Model error: forward models perturbed from the truth by realistic systematics, testing false alarms of h_0 .

Ablations

Remove each term of the utility in turn (no h_0 ; no decay; no inadequacy term but keep h_0 in the belief), replace the auction with independent greedy planners, and vary K . Each ablation maps directly to one hypothesis, so a reader can see which part earns its keep.

Statistics

Run at least 1,000 injected events per scenario with fixed seeds shared across strategies. Compare strategies with paired tests (Wilcoxon signed-rank on per-event metrics), report medians with bootstrap 95% intervals, and correct for multiple comparisons (Holm). Report effect sizes, not only p-values. Tune τ and the weights β, γ on a separate validation set of events; never on the test set.

Real data, honestly used

Real archives cannot answer "what if we had measured something else", because only the measurements actually taken exist. Use real public light curves (for example from ZTF via a broker such as Fink or ALeRCE) for two narrower purposes: to check that the forward models and noise match reality, and for a retrospective replay on densely observed events, where the planner may only choose among epochs that really exist. State this limitation in the paper.

Hardware-in-the-loop testbed

Element	Hardware	Real constraint reproduced
Spacecraft nodes (6–8)	ESP32 boards running belief update, utility and bidding on-device	Limited compute and memory; tests whether the planner fits a microcontroller-class budget
Crosslinks	ESP-NOW radio, gated on and off by the OrbitalNet contact schedule replayed in real time (or accelerated)	Intermittent links, message loss, asynchrony
Clocks	Independent node clocks, synchronised only via messages	Clock drift and ordering errors

Element	Hardware	Real constraint reproduced
Sky oracle and ground	Raspberry Pi serving measurement outcomes from the simulated universe and acting as ground station	Separates truth from the nodes' beliefs
Faults	Power-cycling nodes on a schedule	Node loss and rejoin

The testbed claim must stay modest: it shows the decision layer survives real asynchronous radios and microcontroller compute with results close to simulation. It does not reproduce radiation, real RF link budgets, attitude control or flight software. Report sim-versus-hardware differences for the same seeds as its main result.

Paper: results, limits and validity

Nothing below is demonstrated. These are predictions the experiments are built to confirm or refute.

Expected results

- P beats B0–B2 clearly on time to identification, because they do not use physics likelihoods to pick discriminating bands.
- P versus B3 is the real test. Expect similar performance on modeled classes, better early-time parameter recovery for fast classes (decay term), and much better behaviour on withheld classes (B3 will confidently misclassify them).
- A plausible failure: h_0 raises false alarms when forward models are imperfect (scenario 5). If so, report the false-alarm/detection trade-off curve; that curve is itself useful.
- The auction should approach the oracle at high contact fractions and degrade toward independent greedy at low ones. Where the crossover sits is an unknown worth reporting.

Limitations

- Hypothesis sets are small and hand-chosen; real follow-up faces dozens of classes and subclasses.
- Forward models are simplified; systematics between templates and reality can masquerade as "unmodeled".

- No real spacecraft constellation today carries this payload mix; the constellation is a design study.
- The "none of the above" hypothesis detects inadequacy; it does not invent a new explanation. Say this plainly: the system participates in discovery by preserving evidence and flagging, not by theorising.
- Computing mutual information in real time on microcontrollers may need coarse approximations that change rankings.

Threats to validity

Type	Threat	Mitigation
Internal	Tuning weights on the test events	Separate validation set; fixed seeds; pre-registered metrics in the repository before final runs
Internal	Baselines implemented weakly	Give B3 the same likelihoods and budget as P; tune baselines as carefully as P
Construct	Posterior contraction may not equal "scientific knowledge"	Also report accuracy of recovered physical parameters against injected truth
External	Simulator realism	Validate noise and light curves against real broker data; state which physics is omitted
External	Hardware testbed does not generalise to flight	Limit the claim to decision-layer behaviour under asynchrony and compute limits
Conclusion	Many comparisons	Holm correction; effect sizes; confidence intervals

Reproducibility

Release the simulator, class models (or scripts to fetch them), orbital elements, contact schedules, the seeds, every strategy's code, the analysis notebooks that produce each figure, ESP32 firmware and the wiring diagram. Use an open licence and archive a tagged version on Zenodo with a DOI.

Future work

Scale to larger hypothesis libraries; integrate a real alert stream in shadow mode (planner advises, humans act) with a partner follow-up network; learned amortized design policies to cut onboard compute; D6 as a second paper (inadequacy-driven onboard retention); and, for flight, radiation-tolerant compute, real link budgets and a hosted-payload or rideshare demonstration.

Part V — Where to publish

The best fit is a space-autonomy planning venue first, then an aerospace information-systems or astronomy-methods journal. This is a systems-science contribution with borrowed theory, so top astronomy or top ML venues would likely reject it as out of scope or not novel enough in theory. Check each venue's current call and deadline before committing; the notes below are from general knowledge, not verified this week.

Venue	Type	Why it fits	Review expectation	Verdict
IWPSS (International Workshop on Planning and Scheduling for Space)	Workshop	The JPL/NASA autonomy community, including the people behind POISE and onboard science, reads it	Clear formulation and comparison with operational baselines	First choice for the planning result
i-SAIRAS	Symposium	POISE itself appeared here; space AI and robotics	Systems plus demonstration	Strong alternative
IEEE Aerospace Conference	Conference	Broad aerospace systems; welcomes student work with simulation and testbeds	Engineering completeness	Good, accessible first paper

Venue	Type	Why it fits	Review expectation	Verdict
NeurIPS workshop on Machine Learning and the Physical Sciences	Workshop	Robust experimental design applied to astronomy	Short paper; methodological clarity	Good for an early version of H3 alone
AIAA Journal of Aerospace Information Systems	Journal	Autonomy, decision-making and information systems for aerospace	Full formulation, thorough evaluation	Target journal for the full paper
RAS Techniques and Instruments or Astronomy and Computing	Journal	Astronomy methods and software	Astrophysical realism of classes and noise	Choose if the astronomy side is strongest
Acta Astronautica	Journal	Broad space-systems scope	Mission relevance	Fallback

Realistic path: workshop paper at IWPSS or i-SAIRAS with H1–H3 in simulation, then a journal paper adding H4 and the hardware testbed. Aiming at the journal directly is possible but slower, and an early workshop gets you feedback from the POISE and DSA communities, who are the reviewers you most need to convince.

Part VI — What to do next, week by week

Your existing simulator becomes baseline B1 and the orbital/contact substrate; almost everything new lives in three modules: a sky model, a belief model and a planner. Twelve weeks is realistic for a workshop-quality result if you keep $K = 4$ and one withheld class until week 8.

Weeks	Build	Done when
1	Read every POISE, SHM/SHM-POMDP, DSA/Starling and robust-BED paper in full; write a one-page related-work note	You can state in two sentences what each does not do; if any does your delta, stop and re-scope

Weeks	Build	Done when
1-2	Freeze interfaces in your simulator: feasible-action query per spacecraft per time step; contact-graph query; event injection hook	B1 runs unchanged through the new interfaces
2-3	Sky model: four class forward models with parameter priors; instrument noise from limiting magnitudes	Plots of simulated light curves match published examples by eye and by peak times
3-4	Belief model: grid or particle posterior per class; Bayesian update per measurement	On a single event, posterior converges to the right class given dense data
4-5	Centralized planner with first utility term only (this is B3); also B0 and B2	First paired comparison plot: B0, B1, B2, B3 on 1,000 events
6	Add decay term	Ablation shows effect on fast classes (H2)
7-8	Add h_0 with a Gaussian-process auxiliary model and the inadequacy term; add withheld classes; mode switch	ROC curve for H3; false-alarm curve under model error
9	Decentralize: local beliefs, message merge with measurement IDs, marginal-gain auction; communication sweep	H4 curve: performance versus contact fraction, with oracle and independent-greedy lines
10-11	ESP32 testbed: port belief + bid to firmware, gate ESP-NOW by contact schedule, Pi as sky oracle	Same seeds, sim versus hardware difference reported
11-12	Statistics, figures, write-up; release repository and tag	Draft submitted to your advisor, then venue

The four figures the paper needs

1. Time-to-identification distributions for all strategies (H1).
2. Parameter recovery for fast versus slow classes with and without decay (H2).
3. ROC of $P(h_0)$ on withheld versus modeled events, next to classifier-entropy thresholding (H3).

4. Performance versus contact fraction for oracle, auction and independent greedy (H4).

If you can only finish three weeks' worth, finish figures 1 and 3. Figure 3 is the one no prior system can draw.

What to tell your committee in one sentence

We show that a distributed space observatory that keeps an explicit "none of the above" explanation and accounts for fading signals recovers physics about fast transients that closed-world and reactive planners lose, under identical orbital and communication constraints.

Sources to read and verify

These were located by web search on 24 Sep 2026; abstracts or summaries were read, full texts were not. Read each in full before citing, and confirm author lists.

1. [JPL POISE project page](#) and [POISE i-SAIRAS 2020 paper](#)
2. [Guirguis et al., POMDPs for Autonomous Science Exploration \(SHM-POMDP\), arXiv 2608.03155](#); cites Candela et al., IROS 2017, on Science Hypothesis Maps
3. [NASA: What is Distributed Spacecraft Autonomy?](#) and [DSA Starling 1.0 results, SmallSat 2024](#)
4. [An Information Theory Approach on Deciding Spectroscopic Follow-ups, AJ](#)
5. [Kennamer et al., Active learning with RESSPECT](#)
6. [Real-time active learning with the Fink broker, PASA 2025](#)
7. [NMMA-Astro-COLIBRI, arXiv 2608.17568](#)
8. [Improving robustness to model misspecification in Bayesian experimental design \(OpenReview\)](#)
9. [Metrics for Bayesian optimal experiment design under model misspecification, arXiv 2304.07949](#)
10. [Robust experimental design via generalised Bayesian inference \(GBOED\), arXiv 2511.07671](#)
11. [GRANDMA observations of ZTF/Fink transients](#) and [TOM Toolkit and AEON](#)
12. [On-orbit space AI survey, arXiv 2604.16518](#)
13. [Real-time detection of anomalies in large-scale transient surveys, MNRAS 2022](#)

14. Still to find yourself: Krause and Guestrin on submodularity of information gain; the consensus-based bundle algorithm; ASE/EO-1 autonomous science papers by Chien et al.; the specific light-curve model papers for each class you use.